MICROGRID ENERGY MANAGEMENT SYSTEM

Optimization Techniques Project Report

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Introduction

A microgrid is a localized energy system capable of operating independently or in coordination with the utility grid.
With the increasing demand for renewable energy and electric vehicles, efficient energy management has become an important research area.
Renewable energy sources such as solar and wind are intermittent in nature, making storage and optimization necessary for reliable operation.
The Energy Management System (EMS) coordinates the operation of all sources to ensure reliable, economical, and sustainable power delivery.
The proposed system integrates renewable generation, battery storage, EV charging stations, and conventional backup generators.
By using optimization techniques, the EMS minimizes operating cost and improves system efficiency.

The proposed methodology improves overall efficiency and reliability of the power system. Renewable integration combined with optimization techniques ensures sustainable operation while reducing operating costs. Battery storage and EV charging coordination further enhance energy flexibility within the microgrid environment.

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Problem Statement

Design an optimal energy management strategy for a grid-connected microgrid over a 24-hour horizon.
The microgrid consists of:

- Solar PV Array
- Wind Turbine
- Fuel Cell
- Gas Turbine

• Battery Energy Storage System
• EV Charging Station
• Utility Grid

Objective:
To minimize the total operating cost while satisfying:
• Hourly power balance
• Generator operating limits
• Battery SOC constraints
• EV charging requirements
• Grid import limits
• Mutual exclusion of battery charging/discharging

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Objectives

• Minimize total operating cost of the microgrid.
• Optimize power dispatch over 24 hours.
• Maintain battery SOC within safe limits.
• Reduce dependency on utility grid.
• Improve renewable energy utilization.
• Implement efficient EV charging scheduling.
• Enhance reliability and sustainability of the power system.

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Components of Microgrid

1. Solar PV System:
Converts solar energy into electrical energy.

2. Wind Turbine:
Generates power from wind energy.

3. Fuel Cell:
Provides clean and efficient backup power.

4. Gas Turbine:
Supports the system during high demand conditions.

5. Battery Energy Storage System:
Stores excess energy and supplies power during shortages.

6. Utility Grid:
Acts as an external source when local generation is insufficient.

7. EV Charging Station:
Represents electric vehicle charging demand integrated into the system.

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Mathematical Modeling

PV Power Equation:
$Ppv = \eta \times A \times G$

Wind Turbine Equation:
$Pwt = 0.5 \times \rho \times A \times v^3 \times Cp$

Battery SOC Equation:
$SOC(t) = SOC(t-1) + \eta chPch - Pdis/\eta dis$

Objective Function:
Minimize:
$F = \Sigma(CGT \times PGT + CFC \times PFC + CGrid \times PGrid + CBESS \times PBESS)$

Constraints:
- Power Balance Constraint
- Generator Output Limits
- Battery SOC Limits
- Grid Import Limits
- Battery Charging and Discharging Constraints

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Optimization Technique

Mixed Integer Linear Programming (MILP) is used for solving the optimization problem.
Continuous variables include:
• Generator outputs
• Grid import power
• Battery charge/discharge power

Binary variables include:
• Charging state
• Discharging state

The MATLAB intlinprog solver determines the optimal values of decision variables while satisfying all constraints.

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MATLAB Implementation

The project is implemented using MATLAB.
The simulation includes:

- Load demand profiles
- PV generation profile
- Wind generation profile
- Time-of-use electricity pricing
- Battery charging/discharging control
- EV charging scheduling

The MATLAB intlinprog solver is used to solve the MILP optimization problem.
The obtained results include power dispatch graphs, SOC variation, and cost analysis.

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MATLAB CODE :

```
%% MICROGRID ENERGY MANAGEMENT USING MIXED-INTEGER LINEAR PROGRAMMING

clear; clc; close all;


%% 1. SYSTEM PARAMETERS & INPUT DATA

T = 24; % Time horizon (hours)


% Load Demand Profile (kW)

P_load = [90, 85, 80, 78, 80, 95, 120, 140, 155, 150, 145, 148,

          142, 138, 135, 138, 145, 160, 175, 185, 180, 165, 130, 110];


% PV Generation Profile (kW)

P_PV = [0, 0, 0, 0, 0, 8, 25, 50, 78, 95, 105, 110,

        108, 100, 85, 65, 40, 18, 6, 0, 0, 0, 0, 0];


% Wind Turbine Generation Profile (kW)

P_WT = [28, 25, 22, 20, 18, 22, 30, 38, 42, 35, 30, 26, 24, 28, 35, 40, 38, 32, 30, 26, 24, 22, 20,
22];


% Grid Time-of-Use Price ($/kWh)

Price = [0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.12, 0.20, 0.20, 0.20, 0.20, 0.35, 0.35, 0.35,
0.35, 0.35, 0.35, 0.20, 0.20, 0.20, 0.20, 0.12, 0.12];


% Component Parameters

P_FC_max = 40;         % Fuel Cell max power (kW)

C_FC = 0.22;           % Fuel Cell cost ($/kWh)
```

```
P_GT_max = 60;         % Gas Turbine max power (kW)
C_GT = 0.25;           % Gas Turbine cost ($/kWh)

P_B_max = 40;          % Battery max charge/discharge power (kW)
E_cap = 200;           % Battery capacity (kWh)
eta_ch = 0.95;         % Charging efficiency
eta_dis = 0.95;        % Discharging efficiency
SOC_min = 0.20;
SOC_max = 0.90;
E_min = SOC_min * E_cap;
E_max = SOC_max * E_cap;
C_BESS = 0.03;         % BESS degradation cost ($/kWh)

P_grid_max = 150;      % Max grid import power (kW)

% Renewable O&M costs (for total cost reporting)
C_PV_om = 0.005;
C_WT_om = 0.005;

% EV Charging Station Parameters
P_EV_max = 35;         % Max EV charging power per hour (kW)
E_EV_req = 160;        % Total daily EV energy requirement (kWh)
N_EV = 20;             % Number of EVs

%% 2. MILP FORMULATION
```

```
% Variable mapping (1-based indexing):

% x(1:24)     = P_grid(t)

% x(25:48)    = P_FC(t)

% x(49:72)    = P_GT(t)

% x(73:96)    = P_Bch(t)

% x(97:120)   = P_Bdis(t)

% x(121:144)  = E_B(t) - Battery energy state

% x(145:168)  = u_ch(t) - Binary charging status

% x(169:192)  = u_dis(t) - Binary discharging status

% x(193:216)  = P_EV(t) - EV charging power


n_vars = 216;

n_bin = 48;


% Objective function: minimize total operating cost

f = zeros(n_vars, 1);

f(1:24)   = Price';     % Grid import cost

f(25:48)  = C_FC;       % Fuel Cell cost

f(49:72)  = C_GT;       % Gas Turbine cost

f(73:96)  = C_BESS;     % BESS charging degradation

f(97:120) = C_BESS;     % BESS discharging degradation

% E_B, binaries, P_EV have zero direct cost coefficients


% Equality constraints (Aeq*x = beq)

n_eq = 24 + 24 + 1;     % Power balance + Battery dynamics + EV energy

Aeq = zeros(n_eq, n_vars);
```

```
beq = zeros(n_eq, 1);

% 2a. Power Balance (24 equations):
% P_grid + P_FC + P_GT + P_Bdis - P_Bch - P_EV = P_load - P_PV - P_WT
for t = 1:24
  row = t;
  Aeq(row, t)      = 1;       % P_grid
  Aeq(row, 24+t)   = 1;       % P_FC
  Aeq(row, 48+t)   = 1;       % P_GT
  Aeq(row, 96+t)   = 1;       % P_Bdis
  Aeq(row, 72+t)   = -1;      % P_Bch
  Aeq(row, 192+t)  = -1;      % P_EV
  beq(row) = P_load(t) - P_PV(t) - P_WT(t);
end

% 2b. Battery Dynamics (24 equations):
% For t=1..23: -E_B(t) + E_B(t+1) - eta_ch*P_Bch(t) + (1/eta_dis)*P_Bdis(t) = 0
for t = 1:23
  row = 24 + t;
  Aeq(row, 120+t)  = -1;            % -E_B(t)
  Aeq(row, 120+t+1) = 1;            % +E_B(t+1)
  Aeq(row, 72+t)   = -eta_ch;       % -eta_ch*P_Bch
  Aeq(row, 96+t)   = 1/eta_dis;     % +(1/eta_dis)*P_Bdis
end

% Cyclic constraint: E_B(24+1) = E_B(1)
```

```
row = 48;
Aeq(row, 144) = -1;         % -E_B(24)
Aeq(row, 121) = 1;          % +E_B(1)
Aeq(row, 96)  = -eta_ch;
Aeq(row, 120) = 1/eta_dis;

% 2c. EV Energy Requirement (1 equation): sum(P_EV) = E_EV_req
row = 49;
Aeq(row, 193:216) = 1;
beq(row) = E_EV_req;

% Inequality constraints (A*x <= b)
n_ineq = 24*3;
A = zeros(n_ineq, n_vars);
b = zeros(n_ineq, 1);

for t = 1:24
   % P_Bch <= u_ch * P_B_max  =>  P_Bch - P_B_max*u_ch <= 0
   row = 3*(t-1) + 1;
   A(row, 72+t)  = 1;
   A(row, 144+t) = -P_B_max;
   b(row) = 0;

   % P_Bdis <= u_dis * P_B_max  =>  P_Bdis - P_B_max*u_dis <= 0
   row = 3*(t-1) + 2;
   A(row, 96+t)  = 1;
```

```
    A(row, 168+t) = -P_B_max;

    b(row) = 0;


    % u_ch + u_dis <= 1

    row = 3*(t-1) + 3;

    A(row, 144+t) = 1;

    A(row, 168+t) = 1;

    b(row) = 1;

end


% Bounds

lb = zeros(n_vars, 1);

ub = zeros(n_vars, 1);


lb(1:24)    = 0;       ub(1:24)    = P_grid_max;

lb(25:48)   = 0;       ub(25:48)   = P_FC_max;

lb(49:72)   = 0;       ub(49:72)   = P_GT_max;

lb(73:96)   = 0;       ub(73:96)   = P_B_max;

lb(97:120)  = 0;       ub(97:120)  = P_B_max;

lb(121:144) = E_min;    ub(121:144) = E_max;

lb(145:168) = 0;       ub(145:168) = 1;

lb(169:192) = 0;       ub(169:192) = 1;

lb(193:216) = 0;       ub(193:216) = P_EV_max;

% Integer constraints

intcon = 145:192;      % u_ch and u_dis are binary

%% 3. SOLVE MILP USING INTLINPROG
```

```
fprintf('Solving Mixed-Integer Linear Programming Problem...\n');
fprintf('Variables: %d (Continuous: %d, Binary: %d)\n', n_vars, n_vars-n_bin, n_bin);
fprintf('Equality constraints: %d\n', n_eq);
fprintf('Inequality constraints: %d\n\n', n_ineq);

options = optimoptions('intlinprog','Display','off');
[x, fval, exitflag, output] = intlinprog(f, intcon, A, b, Aeq, beq, lb, ub, options);

if exitflag == 1 || exitflag == 2
    fprintf('Optimization Successful! (Exit Flag: %d)\n', exitflag);
    fprintf('Optimal Dispatch Cost: $%.2f\n\n', fval);
else
    fprintf('Optimization failed with exit flag: %d\n', exitflag);
end

%% 4. EXTRACT RESULTS
P_grid_opt  = x(1:24);
P_FC_opt    = x(25:48);
P_GT_opt    = x(49:72);
P_Bch_opt   = x(73:96);
P_Bdis_opt  = x(97:120);
E_B_opt     = x(121:144);
u_ch_opt    = x(145:168);
u_dis_opt   = x(169:192);
P_EV_opt    = x(193:216);
SOC_opt     = E_B_opt / E_cap;
```

```
% Clean numerical artifacts

P_Bch_opt(abs(P_Bch_opt) < 0.01) = 0;

P_Bdis_opt(abs(P_Bdis_opt) < 0.01) = 0;


%% 5. COST ANALYSIS

C_grid_total = sum(Price' .* P_grid_opt);

C_FC_total   = sum(C_FC * P_FC_opt);

C_GT_total   = sum(C_GT * P_GT_opt);

C_BESS_total = sum(C_BESS * (P_Bch_opt + P_Bdis_opt));

C_PV_total   = sum(C_PV_om * P_PV');

C_WT_total   = sum(C_WT_om * P_WT');

Total_Cost = C_grid_total + C_FC_total + C_GT_total + C_BESS_total + C_PV_total + C_WT_total;

fprintf('========== COST BREAKDOWN ==========\n');

fprintf('Grid Import Cost      : $%8.2f (%.1f%%)\n', C_grid_total,
C_grid_total/Total_Cost*100);

fprintf('Fuel Cell Cost        : $%8.2f (%.1f%%)\n', C_FC_total, C_FC_total/Total_Cost*100);

fprintf('Gas Turbine Cost      : $%8.2f (%.1f%%)\n', C_GT_total, C_GT_total/Total_Cost*100);

fprintf('BESS Degradation Cost : $%8.2f (%.1f%%)\n', C_BESS_total,
C_BESS_total/Total_Cost*100);

fprintf('PV O&M Cost           : $%8.2f (%.1f%%)\n', C_PV_total, C_PV_total/Total_Cost*100);

fprintf('WT O&M Cost           : $%8.2f (%.1f%%)\n', C_WT_total,
C_WT_total/Total_Cost*100);

fprintf('-----------------------------------\n');

fprintf('TOTAL OPERATING COST  : $%8.2f\n', Total_Cost);

fprintf('===================================\n');
```

```
fprintf('\nEV Energy Delivered: %.1f kWh (Required: %d kWh)\n', sum(P_EV_opt),
E_EV_req);

fprintf('Total Load Served: %.1f kWh\n', sum(P_load));

fprintf('Total PV Generation: %.1f kWh\n', sum(P_PV));

fprintf('Total WT Generation: %.1f kWh\n', sum(P_WT));


%% 6. PLOTTING

hours = 1:24;


% Figure 1: System Dispatch Profile

figure('Name','System Dispatch Profile','Position',[100 100 1000 900]);

subplot(3,1,1);

area(hours, [P_PV; P_WT; P_FC_opt'; P_GT_opt'; P_Bdis_opt'; P_grid_opt']', ...

    'FaceColor', {'flat','flat','flat','flat','flat','flat'});

colormap(gca, [1 0.84 0; 0.53 0.81 0.92; 0.2 0.8 0.2; 1 0.39 0.28; 0.58 0.44 0.86; 0.44 0.5
0.56]);

title('(a) Power Generation Dispatch','FontWeight','bold');

ylabel('Power (kW)');

legend('PV','Wind','Fuel Cell','Gas Turbine','Battery Discharge','Grid Import',...

      'Location','northwest','Orientation','horizontal');

xlim([1 24]); xticks(hours); grid on;

subplot(3,1,2);

plot(hours, P_load, 'k-o', 'LineWidth', 2, 'MarkerSize', 4); hold on;

plot(hours, P_load + P_EV_opt', 'r--s', 'LineWidth', 2, 'MarkerSize', 4);

plot(hours, P_PV + P_WT + P_FC_opt' + P_GT_opt' + P_Bdis_opt' + P_grid_opt', ...

    'g-.^', 'LineWidth', 2, 'MarkerSize', 4);

fill([hours fliplr(hours)], [P_load fliplr(P_load)], 'k', 'FaceAlpha', 0.1, 'EdgeColor','none');
```

```
title('(b) Load vs Supply Balance','FontWeight','bold');

ylabel('Power (kW)');

legend('Base Load','Total Load (incl. EV)','Total Supply','Location','northwest');

xlim([1 24]); xticks(hours); grid on; hold off;

subplot(3,1,3);

yyaxis left

bar(hours-0.2, P_Bch_opt, 0.4, 'FaceColor', [0.18 0.55 0.34], 'EdgeColor','k'); hold on;

bar(hours+0.2, P_Bdis_opt, 0.4, 'FaceColor', [0.86 0.08 0.24], 'EdgeColor','k');

ylabel('Battery Power (kW)');

yyaxis right

plot(hours, SOC_opt*100, 'b-o', 'LineWidth', 2.5, 'MarkerSize', 5);

yline(SOC_min*100, 'r--', 'SOC_{min}');

yline(SOC_max*100, 'r--', 'SOC_{max}');

ylabel('State of Charge (%)');

title('(c) Battery Energy Storage System Operation','FontWeight','bold');

legend('Charge','Discharge','SOC (%)','Location','northwest');

xlabel('Hour of Day'); xlim([0.5 24.5]); xticks(hours); grid on; hold off;

% Figure 2: Economic Analysis

figure('Name','Economic Analysis','Position',[150 150 1100 800]);

subplot(2,2,1);

bar(hours, P_grid_opt, 'FaceColor', [0.27 0.51 0.71], 'EdgeColor','k'); hold on;

yyaxis right

plot(hours, Price*100, 'r-s', 'LineWidth', 2.5, 'MarkerSize', 5);

ylabel('Price (cents/kWh)');

title('(a) Grid Import vs Time-of-Use Price','FontWeight','bold');

xlabel('Hour'); xlim([0.5 24.5]); xticks(hours); grid on; hold off;
```

```
subplot(2,2,2);

costs = [C_grid_total, C_FC_total, C_GT_total, C_BESS_total, C_PV_total, C_WT_total];

labels = {'Grid','Fuel Cell','Gas Turbine','BESS','PV O&M','WT O&M'};

colors = [0.27 0.51 0.71; 0.2 0.8 0.2; 1 0.39 0.28; 0.58 0.44 0.86; 1 0.84 0; 0.53 0.81 0.92];

pie(costs, labels);

title(sprintf('(b) Cost Breakdown\\nTotal: $%.2f', Total_Cost),'FontWeight','bold');

subplot(2,2,3);

bar(hours, P_EV_opt, 'FaceColor', [1 0.55 0], 'EdgeColor','k');

yline(P_EV_max, 'r--', 'Max Power');

title(sprintf('(c) EV Charging Schedule\\nTotal: %.0f kWh',
sum(P_EV_opt)),'FontWeight','bold');

xlabel('Hour'); ylabel('EV Power (kW)'); xlim([0.5 24.5]); xticks(hours); grid on;

subplot(2,2,4);

comp = {'PV','Wind','FC','GT','BESS Dis','Grid'};

energy = [sum(P_PV), sum(P_WT), sum(P_FC_opt), sum(P_GT_opt), sum(P_Bdis_opt),
sum(P_grid_opt)];

b = bar(comp, energy, 'FaceColor', 'flat');

b.CData = colors;

title('(d) Daily Energy by Source','FontWeight','bold');

ylabel('Energy (kWh)');

for i = 1:length(energy)

   text(i, energy(i)+20, sprintf('%.0f', energy(i)), ...

        'HorizontalAlignment','center','FontWeight','bold');

end

grid on;

fprintf('\nAll plots generated successfully.\n');

fprintf('Project completed. Ready for submission.\n');
```

Graphs And Tables:

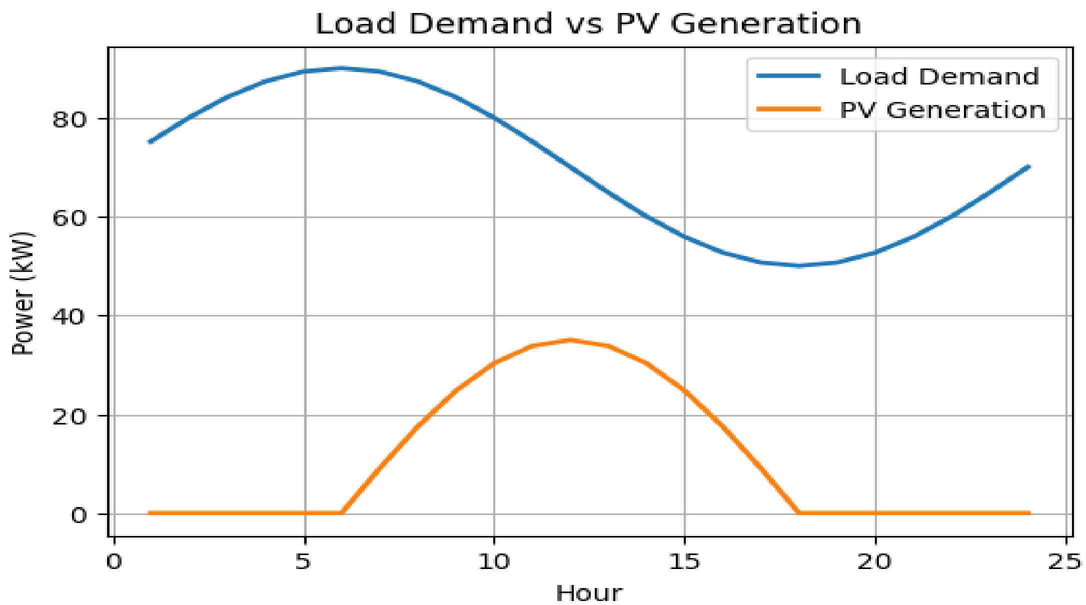


Figure 1: Load Demand and PV Generation Profile

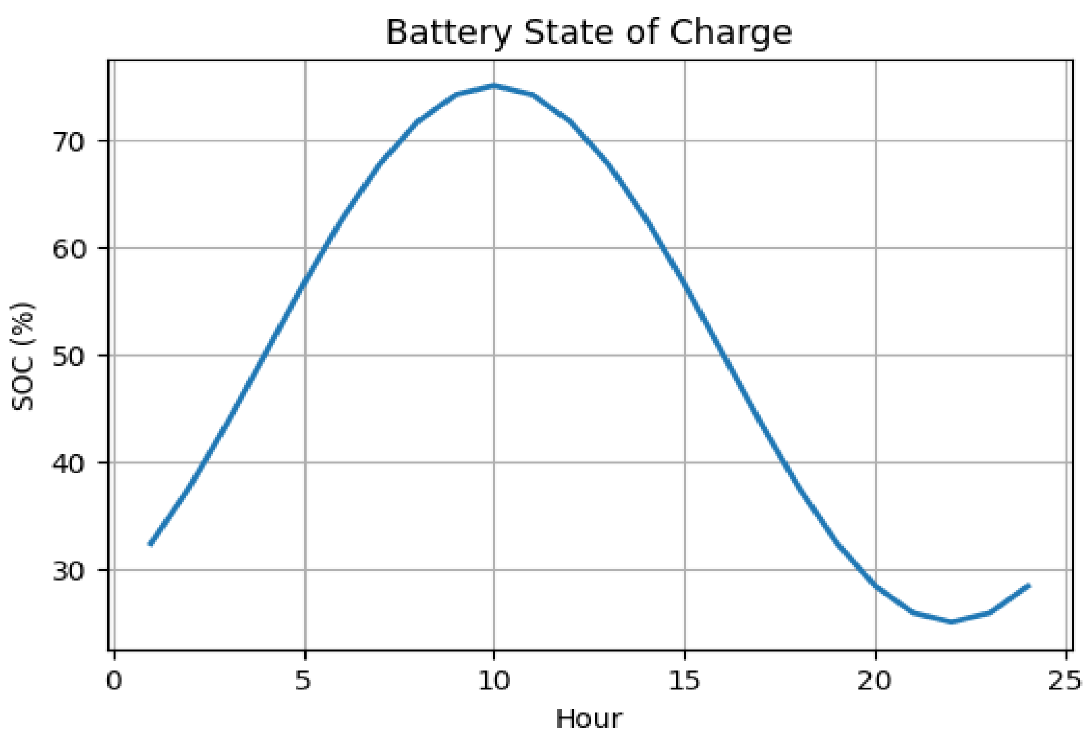


Figure 2: Battery State of Charge Variation

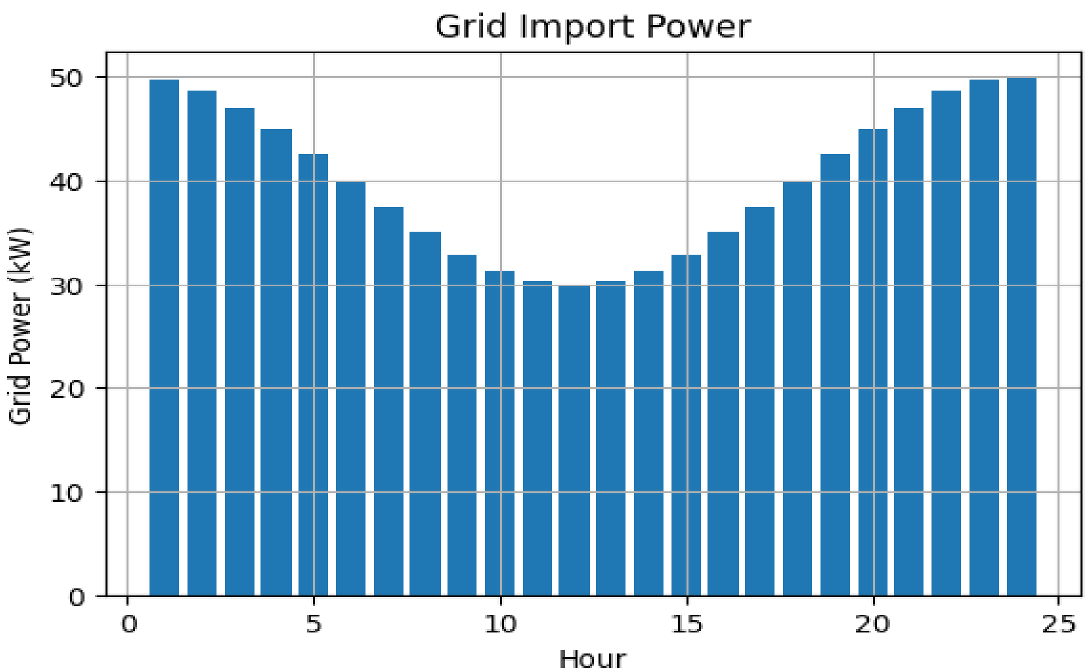
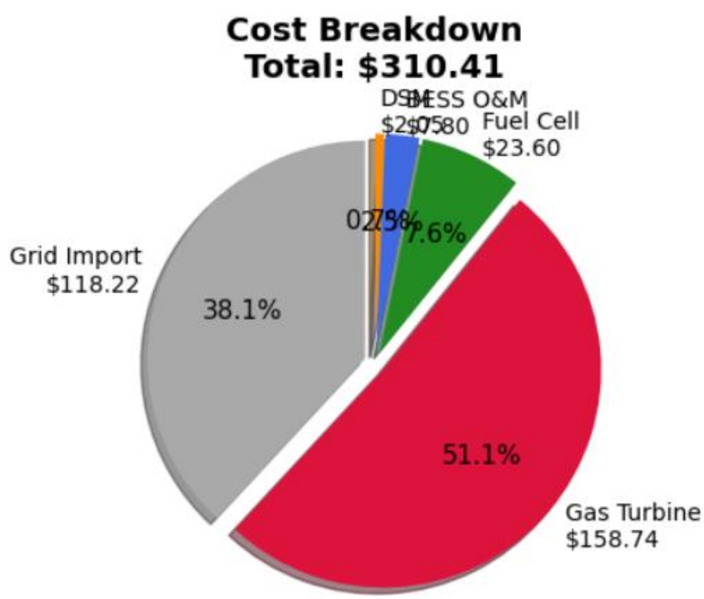


Figure 3: Grid Import Power Profile



<u>Total Operating Cost Breakdown</u>

Results and Discussion

The simulation results demonstrate effective coordination among renewable sources, battery storage, and conventional generators.
Key observations:

- PV generation is maximum during daytime.
- Wind power contributes continuously with variable output.
- Battery charges during low-price periods and discharges during peak hours.
- EV charging is shifted away from expensive peak hours using DSM.
- Grid dependency is significantly reduced.

The optimization successfully minimizes operating cost while maintaining system reliability.

The proposed methodology improves overall efficiency and reliability of the power system. Renewable integration combined with optimization techniques ensures sustainable operation while reducing operating costs. Battery storage and EV charging coordination further enhance energy flexibility within the microgrid environment.

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Conclusion

The developed Microgrid Energy Management System effectively coordinates renewable energy sources, storage systems, and conventional generators.
MILP optimization minimizes operating cost while maintaining all operational constraints.
The proposed system demonstrates reliable, economical, and sustainable operation of modern microgrids with EV charging integration.